



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Course: Full Stack Development

Registration No: 211FA04163

Section: IV – E

Branch: CSE

Submitted To:

Raghav sir

1. Database and Collection Setup

```
> use library
< switched to db library
> db.createCollection("books")
< { ok: 1 }
> db.books.insertOne({'Title': 'To Kill a Mocking Bird', 'Author': 'Harper Lee', 'Publication Year': 1960, 'Genre': 'Fiction', 'ISBN': '978-0446310789'})
< {
  acknowledged: true,
  insertedId: ObjectId('66ea32dbb946dfd5b1953ce2')
}
```

2. Create Operation

In the above I have already inserted one book record into the books collection so now I am inserting only 4 records in this question.

```
> db.books.insertMany(['Title': '1984',
  'Author': 'George Orwell',
  'Publication Year': 1949,
  'Genre': 'Dystopian Fiction',
  'ISBN': '978-0451524935'], {'Title': 'The Da Vinci Code',
  'Author': 'Dan Brown',
  'Publication Year': 2003,
  'Genre': 'Thriller',
  'ISBN': '978-0307474278'}, {'Title': 'A Brief History of Time',
  'Author': 'Stephen Hawking',
  'Publication Year': 1988,
  'Genre': 'Science',
  'ISBN': '978-0553380163'}, {'Title': 'Pride and Prejudice',
  'Author': 'Jane Austen',
  'Publication Year': 1813,
  'Genre': 'Classic Literature',
  'ISBN': '978-0141439518']])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66ea34bab946dfd5b1953ce3'),
    '1': ObjectId('66ea34bab946dfd5b1953ce4'),
    '2': ObjectId('66ea34bab946dfd5b1953ce5'),
    '3': ObjectId('66ea34bab946dfd5b1953ce6')
  }
}
```

3. Read Operation

4. Update Operation

```
> db.books.find({'Author':'Harper Lee'})
< {
  _id: ObjectId('66ea32dbb946dfd5b1953ce2'),
  Title: 'To Kill a Mocking Bird',
  Author: 'Harper Lee',
  'Publication Year': 1960,
  Genre: 'Fiction',
  ISBN: -446309811
}
> db.books.find({'ISBN':978-0141439518})
< {
  _id: ObjectId('66ea34bab946dfd5b1953ce6'),
  Title: 'Pride and Prejudice',
  Author: 'Jane Austen',
  'Publication Year': 1813,
  Genre: 'Classic Literature',
  ISBN: -141438540
}
> db.books.updateOne({'ISBN':978-0141439518},{ $set: {'isAvailable':'true'}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.books.find({'ISBN':978-0141439518})
< {
  _id: ObjectId('66ea34bab946dfd5b1953ce6'),
  Title: 'Pride and Prejudice',
  Author: 'Jane Austen',
  'Publication Year': 1813,
  Genre: 'Classic Literature',
  ISBN: -141438540,
  isAvailable: 'true'
}
```

5. Delete Operation

```
> db.books.deleteOne({'ISBN':978-0446310789})
< {
  acknowledged: true,
  deletedCount: 1
}
```

```
> db.books.deleteMany({'Genre':{'$in':['Science','Dystopian Fiction']}})
< {
  acknowledged: true,
  deletedCount: 2
}
> db.books.find()
< {
  _id: ObjectId('66ea34bab946dfd5b1953ce4'),
  Title: 'The Da Vinci Code',
  Author: 'Dan Brown',
  'Publication Year': 2003,
  Genre: 'Thriller',
  ISBN: -307473300
}
{
  _id: ObjectId('66ea34bab946dfd5b1953ce6'),
  Title: 'Pride and Prejudice',
  Author: 'Jane Austen',
  'Publication Year': 1813,
  Genre: 'Classic Literature',
  ISBN: -141438540,
  isAvailable: 'true',
  Century: 20
}
```